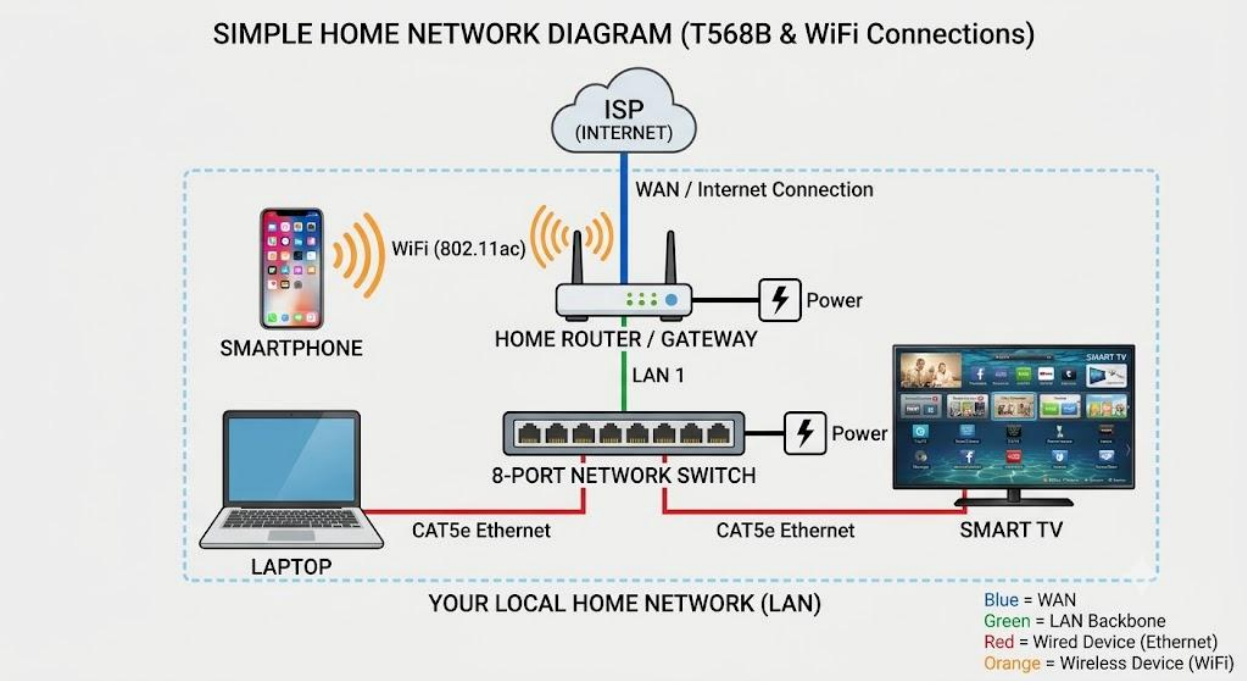
NETWORK DEVICES

1. Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.

Ans:-



2. Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

Ans:-

Network Device	Main Function	Real-World Use Case
Router	Connects different networks and forwards data between them.	At home, a router connects phones and laptops to the internet for watching YouTube videos and using WhatsApp.
Switch	Connects multiple devices within a local area network (LAN).	In a gaming café, a switch connects multiple gaming PCs for LAN gaming and online multiplayer games.
Hub	Connects multiple devices and broadcasts incoming data to all connected ports.	In an older computer lab, a hub may connect multiple computers to share a network.
Access Point (AP)	Provides Wi-Fi connectivity to wireless devices.	At college, an access point allows students to connect their phones and laptops to Wi-Fi for browsing websites and attending online classes.

3. Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, and access point).

Ans:-

Scenario: Ordering Food on Zomato

1. Phone: I open the Zomato app, select food, and place my order. My phone sends the order data over Wi-Fi.

2. Access Point (AP): The access point provides a wireless connection between my phone and the local network.

3. Router: The router forwards my order data from my home network to the internet through my Internet Service Provider (ISP).

4. Switch: A switch forwards data between devices within the network, such as the router and other wired network equipment. Switches may also be used within the ISP or Zomato's data center.

5. Internet and Zomato Server: The order request travels across the internet to Zomato's servers, where it is processed and sent to the restaurant.

6. Response to My Phone: Zomato sends an order confirmation back through the internet, router, and access point to my phone.

4. Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.

Hint: Think about devices like switches, power line adapters, or WiFi extenders.

Ans:-

* Two Ways to Connect a Gaming PC to the Internet

1. Powerline Adapter

- How it works: It uses the existing electrical wiring in the house to carry network data.
- Setup: Connect one powerline adapter to the Wi-Fi router using an Ethernet cable and plug it into a wall socket. Plug the second adapter near the gaming PC and connect it to the PC using another Ethernet cable.
- Use: It provides a wired connection without running a long Ethernet cable across the house.

2. Wi-Fi Extender with an Ethernet Port

- How it works: It receives the router's Wi-Fi signal and extends network coverage. Its Ethernet port lets you connect a wired-only gaming PC.
- Setup: Place the extender where it receives a good Wi-Fi signal, then connect the extender to the gaming PC using an Ethernet cable.
- Use: It allows the gaming PC to access the internet without needing built-in Wi-Fi.